

Constant Time Tree Reachability

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1: function BUILDLISTS( $A$ )
2:    $L_1 \leftarrow$  Empty list
3:    $L_2 \leftarrow$  Empty list
4:   for child  $C$  of  $A$  do
5:      $(T_1, T_2) \leftarrow$  BUILDLISTS( $C$ )
6:      $L_1 \leftarrow$  CONCAT( $L_1, T_1$ )
7:      $L_2 \leftarrow$  CONCAT( $T_2, L_2$ )
8:   end for
9:    $L_1 \leftarrow$  CONCAT( $\{A\}, L_1$ )
10:   $L_2 \leftarrow$  CONCAT( $\{A\}, L_2$ )
11:  return ( $L_1, L_2$ )
12: end function
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Let $I_1(A)$ and $I_2(A)$ be the position of A in the first and second list returned by BUILDLISTS(R) where R is the root of the tree. Then B is an (indirect) child of A if and only if $I_1(B) > I_1(A)$ and $I_2(B) > I_2(A)$.

Proof. First if B is a direct or indirect child of a A then it is clear from lines 9 and 10 that B appears in both lists after A . Thus $I_j(B) > I_j(A)$ for both lists.

Second if B is not a child of A , let U be the first common parent of A and B and C_A and C_B the direct children of U that are parents of A and B respectively. Without loss of generality assume that in the loop on lines 4 to 8 C_A is visited before C_B . Then C_A and all of its children will appear in L_1 before C_B its children. For L_2 the opposite is the case. Hence $I_1(A) < I_1(B)$ and $I_2(A) > I_2(B)$. $\square$